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COMPUTER

CHAPTER 12

Computer Network Architectures

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COMPUTER · CHAPTER 12

Computer Network Architectures

ਕੰਪਿਊਟਰ ਨੈੱਟਵਰਕ ਆਰਕੀਟੈਕਚਰ

IN THIS CHAPTER

- Computer Network Architectures
- Types Of Network Architectures
- Client-Server Architecture
- Peer-To-Peer (P2P) Architecture
- Hybrid Network Architecture
- Transmission Media (Transmission Channel)

COMPUTER NETWORK ARCHITECTURES

A **Network Architecture** refers to the overall design and structure of a computer network. It defines how computers communicate, share resources, and exchange information with each other.

TYPES OF NETWORK ARCHITECTURES

Computer Network Architectures

```
|
├── Client-Server Architecture
├── Peer-to-Peer (P2P) Architecture
└── Hybrid Architecture
```

1. CLIENT-SERVER ARCHITECTURE

In Client-Server Architecture, one or more powerful computers act as **Servers**, while other computers act as **Clients**.

The server provides resources and services, and clients request those resources.

Components

1. Server

A central and powerful computer that:

- Stores data
- Manages resources
- Processes requests
- Provides services to clients

2. Client

A computer or device that requests services from the server.

Working Process

Client Request

↓

Server

(Process Request)

↓

Response Sent

↓

Client

Steps

1. Client sends a request.
2. Server receives the request.
3. Server processes the request.
4. Server sends the response back to the client.

Examples

- Email Servers
- Web Servers
- Facebook
- Instagram
- Online Banking Systems

Advantages

- Centralized management
- Better security
- Easy data backup
- Faster resource sharing

Disadvantages

- Server failure affects the entire network.
- Costly to maintain.

2. PEER-TO-PEER (P2P) ARCHITECTURE

In Peer-to-Peer Architecture, all computers have equal status and power.

There is no central server.

Each computer can act as both:

- Client
- Server

Working

Every computer can:

- Share files
- Share memory
- Share processing power
- Access resources of other computers

Structure

Peer ↔ Peer

↕ ↕

Peer ↔ Peer

Examples

- Bluetooth File Sharing
- BitTorrent

Advantages

- Low cost
- Easy setup
- No central server required

Disadvantages

- Less secure
- Difficult management
- Difficult backup

3. HYBRID NETWORK ARCHITECTURE

Definition

Hybrid Architecture combines the features of:

- Client-Server Architecture
- Peer-to-Peer Architecture

Example

A company may use:

- Central server for official data
- Peer-to-peer sharing among employees

Client-Server vs Peer-to-Peer

Feature	Client-Server	Peer-to-Peer
Server	Present	Absent
Control	Centralized	Decentralized

Feature	Client-Server	Peer-to-Peer
Security	High	Lower
Cost	High	Low
Management	Easy	Difficult
Examples	Facebook, Email	Bluetooth Sharing

TRANSMISSION MEDIA (TRANSMISSION CHANNEL)

A Transmission Medium (Channel) is the path through which data travels from sender to receiver.

When you send a WhatsApp message, the data travels through transmission media.

TYPES OF TRANSMISSION MEDIA

Transmission Media

│

├── Guided (Wired)

│ ├── Twisted Pair Cable

│ ├── Coaxial Cable

│ └── Optical Fiber

│

└── Unguided (Wireless)

├── Radio Waves

├── Infrared

└── Microwaves

GUIDED TRANSMISSION MEDIA (WIRED MEDIA)

Guided Media uses physical cables to carry signals.

Characteristics

- High Speed
- More Secure
- Less Interference
- Suitable for Short and Medium Distances

1. TWISTED PAIR CABLE

Twisted Pair Cable consists of pairs of insulated copper wires twisted together.

Why Twisted?

Twisting reduces:

- Noise
- Electromagnetic interference
- Crosstalk

Types of Twisted Pair Cable

A. UTP (Unshielded Twisted Pair)

- No metallic shielding
- Cheaper
- Lightweight
- Easy installation

Examples

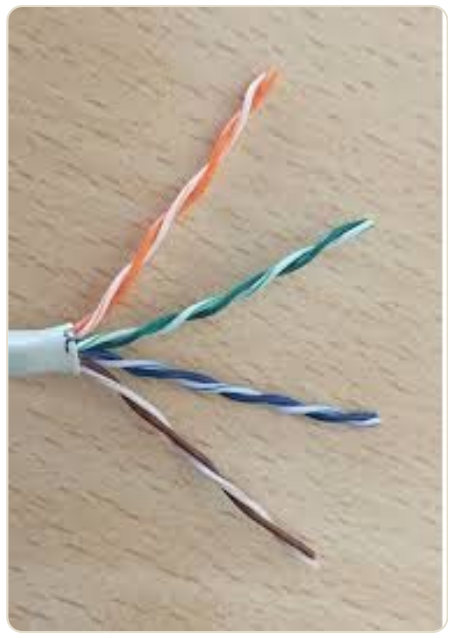
- Telephone lines
- LAN cables

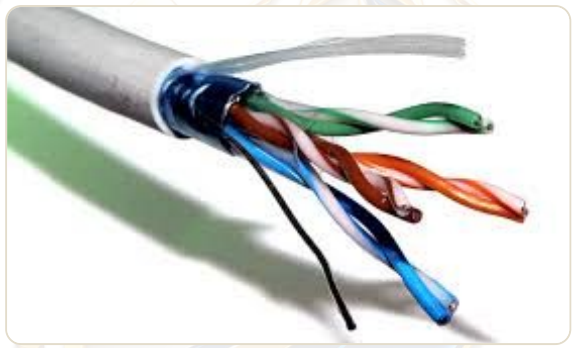
B. STP (Shielded Twisted Pair)

- Metallic shielding present
- Better protection from interference
- Used in high-speed networks

Examples:

- CAT3
- CAT5
- CAT6
- CAT7





2. COAXIAL CABLE

A cable consisting of:

Outer Plastic Cover

↓

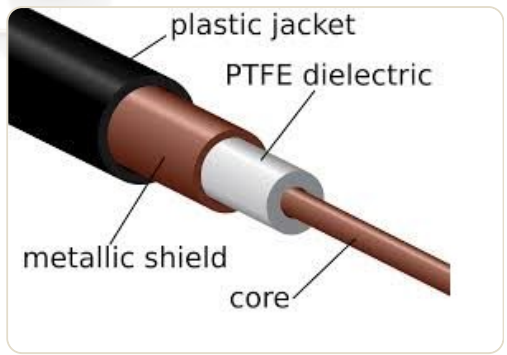
Metal Shield

↓

Insulation

↓

Copper Core



Characteristics

- Better protection from interference
- Longer transmission distance
- Higher bandwidth than twisted pair

Uses

- Cable TV

- Analog Television
- Broadband Internet

3. OPTICAL FIBER (FIBER OPTIC CABLE)

A fibre optic cable is a specialised medium designed to transmit data by converting information into pulses of light. These light signals travel through extremely thin, flexible strands made of either glass or high-quality plastic, known as optical fibres. Each strand acts like a tiny, transparent pipe The technology leverages the principle of total internal reflection, ensuring that the light pulses remain contained within the fibres, which makes fibre optics highly efficient for internet, telecommunication, and networking applications.

Characteristics

- Very High Speed
- High Bandwidth
- No Electromagnetic Interference
- Long Distance Communication

Structure

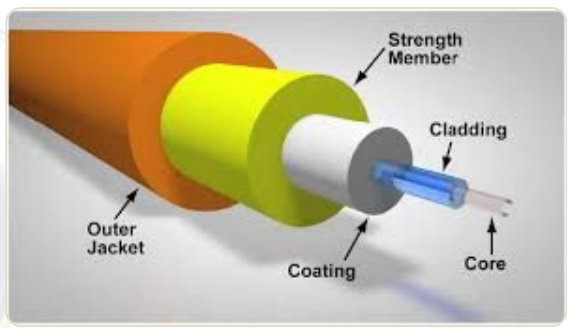
Fiber Optic Cable

│

├── Core

├── Cladding

└── Jacket



Components of Fiber Optic Cable

1. Core

- Central part of the fiber
- Carries light signals

2. Cladding

- Surrounds the core
- Reflects light back into the core

3. Jacket

- Outer protective covering
- Protects against damage

Advantages

- Fastest transmission medium
- High security
- Long-distance communication

Uses

- Internet Backbone
- Undersea Communication Cables
- Telecommunication Networks

Exam Point

Fiber Optic Cable transmits data using Light Signals.

THROUGHPUT

Throughput is the actual amount of data successfully transmitted through a network during a specific time period.

Other Names

- Effective Data Rate
- Payload Rate

Example

If a network can theoretically transfer 100 Mbps but actually transfers 80 Mbps, then throughput is 80 Mbps.

PARITY CHECK

Parity Check is an error detection technique used during data transmission.

A parity bit is added to data to check whether data has been transmitted correctly.

UNGUIDED TRANSMISSION MEDIA (WIRELESS MEDIA)

Unguided Media transmits electromagnetic waves without physical cables.

Air acts as the transmission medium.

Characteristics

- Wireless communication
- Easy installation
- Suitable for long distances

TYPES OF UNGUIDED MEDIA

1. RADIO WAVES

Radio waves are electromagnetic waves that travel in all directions used for long distance communication

Frequency Range

3 KHz to 1 GHz

Examples

- FM Radio
- AM Radio
- Television Broadcasting

Exam Point

Radio Waves are Omnidirectional.

2. INFRARED (IR)

Infrared is wireless communication technology used for short distances.

Frequency Range

300 GHz to 400 THz

Characteristics

- Short-range communication
- Cannot pass through walls
- Requires line-of-sight

Examples

- TV Remote Control
- Air Conditioner Remote
- Mobile Data Transfer

Bluetooth actually uses **Radio Waves (2.4 GHz)**, not Infrared.

Exam Point

TV Remote = Infrared Technology

3. MICROWAVES

Microwaves are high-frequency electromagnetic waves used for long-distance communication.

Frequency Range

1 GHz to 1000 GHz

Characteristics

- Unidirectional
- Requires antenna alignment
- Uses Line-of-Sight communication

Types of Microwaves

A. Terrestrial Microwave

Communication between two ground-based antennas.

Structure

Tower A ↔ Tower B

Uses

- Mobile Networks
- Television Broadcasting
- Internet Backbone

B. Satellite Microwave Communication

Communication using satellites orbiting the Earth.

Working

Earth Station

↑

Satellite

↓

Earth Station

Advantages

- Global coverage
- Reliable communication
- Long-distance communication

Examples

- Live TV Broadcast
- GPS
- Mobile Communication
- Weather Forecasting

TRANSMISSION MODES

Transmission Mode refers to the direction of data flow between two communicating devices in a network.

In simple words, transmission mode determines **how data is sent and received between devices**.

It is also known as:

- Communication Mode
- Data Communication Mode

Example

When you make a phone call, send a message, or watch TV, data is being transferred using a particular transmission mode.

TYPES OF TRANSMISSION MODES

There are three types of transmission modes:

Transmission Modes

│

├── Simplex Mode

├── Half-Duplex Mode

└── Full-Duplex Mode

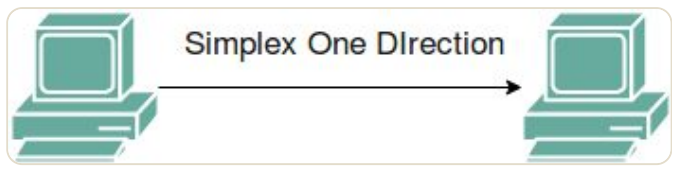
1. SIMPLEX MODE

In **Simplex Mode**, communication takes place in only one direction.

One device can only transmit data, while the other device can only receive data.

Characteristics

- One-way communication
- No feedback from receiver
- Entire channel capacity is used in one direction
- Simplest communication mode



Examples

- Keyboard → Computer
- Traditional Monitor ← Computer
- Television Broadcasting
- Dish TV
- Radio Broadcasting

Real-Life Example

When watching television, the TV station sends signals to your TV, but your TV cannot send signals back.

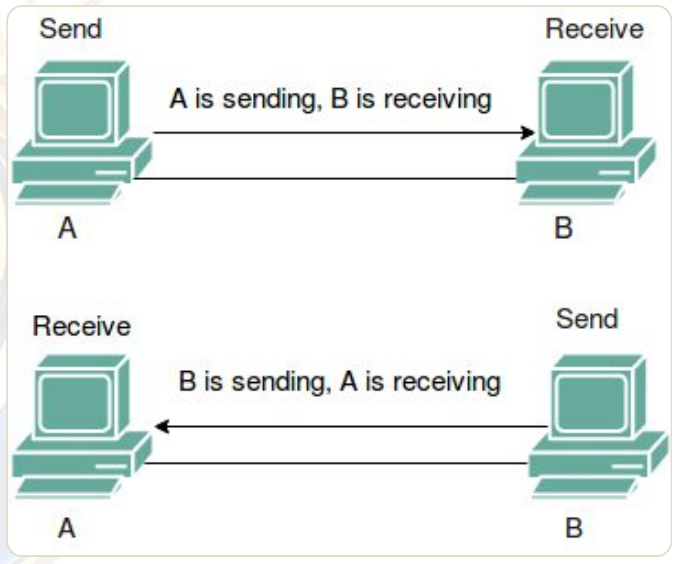
2. HALF-DUPLEX MODE

In **Half-Duplex Mode**, both devices can transmit and receive data, but **not at the same time**.

When one device is sending, the other device can only receive.

Characteristics

- Two-way communication
- One direction at a time



Examples

- Walkie-Talkie
- Wireless Radio Communication

Real-Life Example

When using a walkie-talkie, one person speaks while the other listens. After finishing, the other person responds.

Exam Point

Half-Duplex = Both Directions, But One at a Time

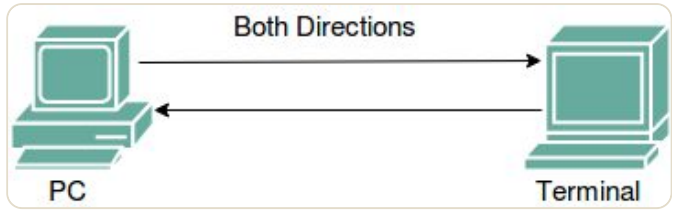
3. FULL-DUPLEX MODE

In **Full-Duplex Mode**, both devices can transmit and receive data simultaneously.

Communication occurs in both directions at the same time.

Characteristics

- Two-way communication
- Simultaneous transmission and reception
- Faster communication
- Most efficient transmission mode



(Both directions simultaneously)

Examples

- Telephone Network
- Mobile Phone Calls
- Video Calls
- Internet Voice Calls (VoIP)

Real-Life Example

During a mobile phone conversation, both people can speak and listen at the same time.

Exam Point

Full-Duplex = Simultaneous Send and Receive

SIMPLEX VS HALF-DUPLEX VS FULL-DUPLEX

Feature	Simplex	Half-Duplex	Full-Duplex
Communication Direction	One Way	Two Way	Two Way
Simultaneous Communication	No	No	Yes
Feedback Possible	No	Yes	Yes
Channel Usage	One Direction Only	One Direction at a Time	Both Directions Simultaneously
Speed	Low	Medium	High
Complexity	Simple	Moderate	Complex
Examples	Keyboard, TV	Walkie-Talkie	Telephone, Mobile Call

REAL-WORLD EXAMPLES TABLE

Device / Technology	Transmission Mode
Keyboard	Simplex
Monitor	Simplex
Television	Simplex
Radio Broadcast	Simplex
Walkie-Talkie	Half-Duplex
Police Wireless Set	Half-Duplex
Telephone	Full-Duplex
Mobile Phone	Full-Duplex
Video Calling	Full-Duplex

IMPORTANT EXAM FACTS

Simplex Mode

- Data flows in one direction only.
- Receiver cannot send data back.
- Entire bandwidth is used in one direction.

Half-Duplex Mode

- Communication in both directions.
- One device communicates at a time.
- Commonly used in walkie-talkies.

Full-Duplex Mode

- Simultaneous transmission and reception.
- Most efficient communication mode.
- Used in telephones and mobile phones.

QUICK REVISION NOTES

- ✓ Client requests service.
- ✓ Server provides service.
- ✓ P2P has no central server.
- ✓ Hybrid = Client-Server + P2P.
- ✓ Guided Media = Wired.
- ✓ Unguided Media = Wireless.
- ✓ UTP = No Shield.
- ✓ STP = Shielded.
- ✓ Coaxial = Cable TV.
- ✓ Fiber Optic = Light Signals.
- ✓ Throughput = Actual Data Transfer.
- ✓ Parity Bit = Error Detection.
- ✓ Radio Waves = Omnidirectional.
- ✓ Infrared = TV Remote.
- ✓ Microwaves = Line of Sight.
- ✓ Satellite = Global Communication.

MOST IMPORTANT FACTS FOR EXAMS

1. Client-Server Architecture is centralized.
2. Peer-to-Peer Architecture is decentralized.
3. Hybrid combines both architectures.
4. Twisted Pair Cable is the most commonly used networking cable.
5. UTP has no shielding.
6. STP has metallic shielding.
7. Coaxial Cable is used in Cable TV.
8. Fiber Optic uses light signals.
9. Throughput means actual transmission rate.
10. Parity Bit detects transmission errors.
11. Radio Waves are omnidirectional.
12. Infrared is used in TV remotes.
13. Microwaves require line-of-sight communication.

14. Satellite communication provides worldwide coverage.